

1. Bubblesort is a popular sorting algorithm that works by repeatedly swapping adjacent elements that are out of order.

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BubbleSort(A,n)
1. for i=1 to n-1
2.     for j=n to i+1
3.         if A[j]<A[j-1]
4.             swap A[j] and A[j-1]
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- (a) Prove that the resulting array contains a rearrangement of the elements in the original array.
 - (b) State precisely a loop invariant for the for loop in lines 2-4, and prove that this loop invariant holds.
 - (c) Using the termination condition of the loop invariant proved in the previous question, state a loop invariant for the for loop in lines 1-4 that allows you to complete the correctness argument of the algorithm.
 - (d) What is the worst-case running time of BubbleSort?
2. An inversion in an array $A[1..n]$ is a pair of indices (i, j) such that $i < j$ and $A[i] > A[j]$. The number of inversions in an n -element array is between 0 (if the array is sorted in non-decreasing order) and $\binom{n}{2}$ (if the array is sorted in non-increasing order).
 - (a) What is the relationship between the running time of insertion sort and the number of inversions in the input array?
 - (b) Describe and analyze an algorithm to count the number of inversions in an n -element array in time proportional to $n \log n$. Hint: Modify merge sort.
3. An interval I is a set of consecutive positive integers $\{x, x+1, \dots, y\}$ represented as $[x, y]$. Two intervals $I_1 = [x_1, y_1]$ and $I_2 = [x_2, y_2]$ are said to overlap if $\{x_1, x_1+1, \dots, y_1\} \cap \{x_2, x_2+1, \dots, y_2\} \neq \emptyset$. Two overlapping intervals $I_1 = [x_1, y_1]$ and $I_2 = [x_2, y_2]$ may be merged into one interval $I = [v, z]$ where $v = \min\{x_1, x_2\}$ and $z = \max\{y_1, y_2\}$. Write a program to merge overlapping intervals in a given set of n intervals in time proportional to $n \log n$.
4. Suppose you are given a stack of n pancakes of different sizes. The objective is to sort the pancakes so that smaller pancakes are on top of larger pancakes. The only operation that you can perform is a flip where you insert a spatula under the top k pancakes for some integer $1 \leq k \leq n$ and flip them all over. Describe an algorithm to sort an arbitrary stack of n pancakes using at most x flips where x is proportional to n . Exactly how many flips does your algorithm perform in the worst case?